

Fall-Semester-2025-2026 (1)

Course code :- BMAT1052

Course Name :- Discrete Mathematics And Graph Theory.

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Tutorial - 4
(Module - 2)

Que-1 Let R be the Additive group of real numbers and let R^+ be the multiplicative group of positive real numbers.

Define: $f: R \rightarrow R^+$ by $f(x) = e^x$

for all $x \in R$. Prove that f is a Homomorphism.



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Solve → The function $f: R \rightarrow R^+$

$f: (G_1 \rightarrow G_2)$ b/w two groups

a group homomorphism. $a, b \in G_1$.

$$f(a+b) = f(a) \cdot f(b)$$

where $(+)$ is the operation in G_1 and $(\cdot)$ is the operation in G_2 .



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Identifying the groups and their operations ②

$= (\mathbb{R}, +)$: Operation in Addition.

$= (\mathbb{R}^+, \times)$: Operation in Multiplication.

$$f(x+y) = f(x) \times f(y)$$

for all $x, y \in \mathbb{R}$.

Step-3 Substitute the given function :-

$$f(x+y) = e^{x+y}$$

$$f(x) \cdot f(y) = (e^x) \cdot (e^y) = e^{x+y}$$

→ on comparison :-

$$f(x+y) = e^{x+y} = e^x \cdot e^y = f(x) \cdot f(y)$$

the Homomorphic property is satisfy ($x, y \in \mathbb{R}$)

① $(\mathbb{R}, +)$:- $f(0) = e^0 = 1$

② $(\mathbb{R}, \times)$ in $-x$:- $f(-x) = e^{-x} = \frac{1}{e^x} = [f(x)]^{-1}$
 showing Inverse also Correspondence property.

Therefore, $f(x) = e^{2x}$ is a group Homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}^+, \times)$ also

Ques-2.
number
multiplication
with
by
 i



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Let T be the additive group of real numbers. Let $f: \mathbb{R} \rightarrow T$ be the map defined by $f(\theta) = \cos \theta + i \sin \theta$. Prove that f is a group homomorphism.

Solve

Show $f: \mathbb{R} \rightarrow T$ given by
 $f(\theta) = \cos \theta + i \sin \theta$ is a
group homomorphism.

Here $T = \{z \in \mathbb{C} : |z| = 1\}$ with
multiplication, domain $(\mathbb{R}, +)$.

Using Euler property :-

$$\rightarrow \cos \theta + i \sin \theta = e^{i\theta}, \text{ Then for } \theta \in \mathbb{R}$$

$$f(s+\theta) = e^{i(s+\theta)} = e^{is} \cdot e^{i\theta} = f(s)f(\theta)$$

Therefore f preserves the operation
 $\rightarrow f$ is a group homomorphism.

Identity: $f(0) = 1$. Inverse
 $f(-\theta) = e^{-i\theta} = (e^{i\theta})^{-1}$

Que-3

Quo:

Then

Solve :-

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is a homomorphism of a (4)
 H is a subgroup of G ,
 $f(H)$ is a subgroup

f' be a homomorphism

prove $f(H)$ is a subgroup of G' .

Proof :-

① nonempty :- $e_G \in H$ because H is a
subgroup. Then $f(e_G) = e_{G'} \in f(H)$

so $f(H)$ is nonempty.

② Closure Under Product :- take $a, b \in f(H)$

Then $a = f(h_1)$, $b = f(h_2)$ with
 $h_1, h_2 \in H$. since f is a
homomorphism.

$$ab = f(h_1)f(h_2) = f(h_1h_2)$$

But $h_1h_2 \in H$ (subgroup), so $ab \in f(H)$

③ Answer



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Let $a \in$
 $n \in$

Since

All $g \in$

inverse :-

⑤

$f(n)$ for some

$$a^{-1} = (f(n))^{-1} = f(n^{-1})$$

$$-1 \in f(n)$$

conditions hold

G' . Therefore,

$f(n)$ is a subgroup of G' .

Ques-4 :- find the codeword generated by a parity check matrix.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

when the encoding function

$$e: B^3 \rightarrow B^6$$

Solve The parity check Matrix can be written as :-

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

So expect $2^n = 2^3 = 8$ codewords.

Let C



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C_4, C_5, C_6

⑥

① C_1

② C_1

③ $C_1 +$

few

from

0

$+C_4$

$+C_4$

$$C_2 + C_4 = C_3 + C_5 \Rightarrow C_2 + C_4 + C_5$$

from ③ $C_1 + C_2 + C_3 + C_6 = 0$

Substitute $C_1 = C_2 + C_4$ and C_3

$$(C_2 + C_4) + C_2 + (C_2 + C_4 + C_5) + C_6 = 0$$

$$C_2 + C_5 + C_6 = 0 \Rightarrow C_2 = C_5 + C_6$$

Now Substituting Back :-

$$C_2 = C_5 + C_6$$

$$C_1 = C_2 + C_4 = C_4 + C_5 + C_6$$

$$C_3 = C_4 + C_6$$

$$C_4 = C_4, \quad C_5 = C_5 \text{ and } C_6 = C_6$$

→ All final keywords are :-

000000, 001011, 011110, 010101, 101100

100111, 110010, 111001

Ans
2

Que-5 for a $(6,3)$ matrix G is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



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generated ⑦
row :-

(i) construct the check matrix :-
Solve $\Rightarrow$ for a Sym for $G = [I_k | P]$

$$H = [P^T | I_{n-k}]$$

Compute $P^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \text{parity check matrix.}$$

(ii) find all codewords generated by G .

$\rightarrow$ All codewords are $2^n = 2^3 = 8$ key codewords

$$G \cdot e(000) = 000000$$

$$G \cdot e(001) = 001110$$

$$G \cdot e(010) = 010011$$

$$G \cdot e(011) = 011101$$

$$G \cdot e(100) = 100101$$

$$G \cdot e(101) = 101011$$

$$G \cdot e(110) = 110110$$

$$G \cdot e(111) = 111000$$

The codewords are

000000, 001110, 010011

011101, 100101, 101011

110110, 111000 $\underline{\text{Ans}}$

(iii) Min

→ Compute
non 2



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Distance :-

(3)

Distance weight of each
d.

(iii) Euro

le :-

→ A codeword with distance d_{min}
detects upto $d_{min}-1$ errors

$$3-1 = 2 \text{ errors}$$

(v) Errors correctable :-

→ A code corrects upto $\left[\frac{d_{min}-1}{2} \right]$ errors.

$$\frac{3-1}{2} = \frac{2}{2} = (1) \rightarrow \text{error can be corrected.}$$

(vi) Received word $r = 100011$. Find the error and correct.

$$z) S = Hr^T \text{ mod } 2 \rightarrow r = (1, 0, 0, 0, 1, 1)^T$$
$$S = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$r \Rightarrow 100011$ Now corrected ($r = 101011$)

plus
2

Que-6 find
by the ϵ
 $e: B^3 \rightarrow$
parity



010
001

is Generated
on
spect to the

Solve

$$H = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Let $C = (C_1, \dots, C_6)$

① $C_3 = 0$

② $C_2 + C_4 = 0$

③ $C_1 + C_2 + C_3 + C_6 = 0$

$C_4 = C_5 + C_6$

$C_1 = C_2 + C_4 = C_2 + C_5 + C_6$

$C_3 = 0$

Total Codewords
② $2^n = 2^3 = 8$
Codewords

The final total 8 codewords are :-

$\{000000, 000011, 010101, 010110, 100101, 100110, 110000, 110011\}$

Ans

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Discrete - Mathematics

Tutorial - 6 (Module - 3)

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Ques-1 Suppose that a person deposits \$10,000 in a savings account at a bank yielding 11% per year with interest compounded annually, how much will be in the account after 30 years

Solve - $A = P \cdot (1 + r)^t$

$$P = \$10000$$

$$r = 11\% = 0.11$$

$$t = 30 \text{ years}$$

$$A = 10000 (1 + 0.11)^{30}$$

$$A = 10000 (1.11)^{30}$$

$\rightarrow (1.11)^{30}$ and then multiply by 10000

$$A = 10000 (22.892296)$$

$$A = 228922.96 \text{ after 30 years}$$

Ans



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Ques-2 A comp
of decimal
it contain
for instar
whereas
let a_n
codeword

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consider a string (1)
and codeword if
number of 0 digit.

869 is valid,

is not valid.

der of valid n -digit
recurrence relation

for a_n .

Solve - Each Digit can be 0-9 (10 possible)

$$a_{n+1} = 9a_n + b_n$$

$$b_{n+1} = 9b_n + a_n$$

Since, $a_n + b_n = 10^n$, we can eliminate b_n :

$$a_{n+1} = 9a_n + (10^n - a_n) = 8a_n + 10^n$$

Recurrence :- $a_{n+1} = 8a_n + 10^n$, $a_1 = 9$ Ans

Ques-3 Suppose that the number of Bacteria in
a colony triples every hour. (a) Set up a
recurrence relation for bacteria after (n) hours
have elapsed. (b) If 100 bacteria are used
to begin a new colony, how many bacteria
will be in the colony in 10 hours?

Solve - Bacteria
at $N_t =$

(a) Each

(b) $N_0 = 10$

Then, after 10 hours :-

$$N_{10} = 100 \times 3^{10} = 100 \times 59049$$

$$N_{10} = 5904900$$

$\Rightarrow 5.9 \times 10^6$ Bacteria Ans

Que-4 How many solutions does the equation
 $x_1 + x_2 + x_3 + x_4 = 17$, where x_1, x_2, x_3
and x_4 are non-negative numbers?

Solve - Equation :- $x_1 + x_2 + x_3 = 11$

Number of non-negative integer
shown solutions are :-

$$\binom{11+3-1}{3-1} = \binom{13}{2} = 78$$

$\Rightarrow 78$ solutions Ans



hour :-
t hours

(12)

us :-

Que-5 How many solutions are there to the equation $x_1 + x_2 + x_3 + x_4 = 17$, where x_1 is negative and x_2, x_3, x_4 are non-negative.



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Solve $\therefore x_1 = -z$

$$\text{Count} = \binom{20}{3} = 1140$$

2) 1140 solutions Ans

Que-6 How many solutions are there to the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 21$, where $x_i = 1, 2, 3, 4, 5$ is a non-negative integer such that (a) $x_1 \geq 1$ (b) $x_1 \geq 2$ for $P = 1, 2, 3, 4, 5$

Solve $\therefore$

$$y_1 = x_1 - 2, \quad y_2 = x_2 - 1$$

$$y_3 = x_3 - 4, \quad y_4 = x_4, \quad y_5 = x_5$$

All $y_i \geq 0$

$$\text{Then } (y_1 + 2) + (y_2 + 1) + (y_3 + 4) + y_4 + y_5 = 21$$

$$y_1 + y_2 + y_3 + y_4 + y_5 = 14$$

$$\binom{14+5-1}{5-1} = \binom{18}{4} = 3060$$

2) 3060 solutions Ans

Que-7 How many
8 either start
with the

Solve - 0 let



length of length (14)
1 or end
0?

of valid binary
of length n with
five 0s.

So Recurrence

$$a_n = a_{n-1} + a_{n-2}$$

with base case,

$$a_1 = 2 ("0", "1")$$

$$a_2 = 3 ("01", "10", "11")$$

that the fibonacci series sequence method

$$a_3 = 5, a_4 = 8, a_5 = 13, a_6 = 21, a_7 = 34$$

$$a_8 = 55$$

Therefore, 55 Binary strings of length 8
with no consecutive 0s.

Que-8 There are 315 students at a college
who have taken a course in Calculus,
212 who have taken a course in discrete
and 186 who have taken course in
both Calculus and discrete maths.

How many st
In either
Solve :- / C.V.



taken a course (15)
Discrete Maths?

$$\begin{aligned} & A + 10 - (C \cap A) - (C \cap D) \\ & 1 + (C \cap D) \\ & + 260 - 120 - 90 - 80 + 50 \\ & 0 + 50 \end{aligned}$$

$\therefore$ 575, Students took at least
one course Ans

Que - 9 Find the number of positive
numbers not exceeding 100 that are
not are divisible by 5 or 7?

Solve :- number of integers divisible
by 5 $\Rightarrow \left\lceil \frac{100}{5} \right\rceil = 20$

Number divisible by 7

$$\Rightarrow \left\lceil \frac{100}{7} \right\rceil = 14$$

Both by 5 or by 7 in 35

$$\left\lceil \frac{100}{35} \right\rceil = 2$$

$$\begin{aligned} \text{Total} &= \text{Div}(5) + \text{Div}(7) - \text{Div}(5 \text{ and } 7) \\ &= 20 + 14 - 2 \\ &= 32 \text{ Ans } \end{aligned}$$

Que-12 Let $S = \{2, 4, 6, 12, 20, 24\}$
by divisibility

Diagram of
maximal and
minimal elements

Solve :- Let $S = \{2, 4, 6, 12, 20, 24\}$

Covering P

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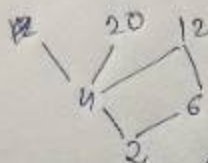
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is ordered (16)

Hasse
Diagram of the
element of S .

ordered
by.

$(2, 6), (4, 12),$
 $(4, 20), (6, 12)$



(Hasse Diagram)

→ Minimal = $\{2\}$

→ Maximal = $\{12, 20\}$ Ans

Que-14 How many different strings can be
made from the letters MISSISSIPPI
using all the letters?

Solve :- letters are multiplied
in (MISSISSIPPI)

$M=1, I=4, S=4, P=2$ {Total = 11}



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Distinct permutations of
111224 :-

(17)

$$\frac{111}{1!4!2!4!}$$

$$\frac{111}{1! \times 4! \times 4! \times 2!} = 34,650$$

∴ 34,650 Distinct strings du

Ques-15 How many different strings can be made from the letter O R O N O, using some or all of the letters?

Solve :- (ORONO) ⇒ O=3, R=1, N=1

By formula :- $\frac{(a+r+n)!}{a!r!n!}$

length 1 : 3 strings
length 2 : 7 strings
length 3 : 13 strings
length 4 : 20 strings
length 5 : 20 strings

Sum ⇒

$$3+7+13+20+20=63$$

∴ Total 63
Distinct strings
(using some or
all letters)

du

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Discrete - Mathematics

Tutorial - 5 (Module - 3)

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Ques-1 In a class of 50 students, play football, and 16 students hockey. It is found that 10 play Both games. Find the number of students who play neither.

Solve :-
Football = 20
Hockey = 16

Football and Hockey = 10
Both

• Number who play atleast one

$$\Rightarrow 20 + 16 - 10 = 26$$

Now, $\Rightarrow$ Total - (at least one)

(neither) $\Rightarrow 50 - 26$

$\Rightarrow 24$ Students $\frac{24}{2}$



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Que-2 How many
select five
number, to
trip to a



Solve - Choose

$$\binom{10}{5}$$

are there to
from a 10-
to make a
to other school?

$$10 \Rightarrow \binom{10}{5}$$

$$252$$

$\therefore 252$ ways Ans

Que-3 Find the number of bit strings of
length 10 that either begin with
1 or end with 0.

Solve :- Strings that fix with 1 : remain $9 = 2^9$

Strings that end with 0 : remain $9 = 2^9$

Strings that end both begin with 1 AND

end with 0 : fix both $\rightarrow$ remain $8 = 2^8$
bits

$$\Rightarrow 2^9 + 2^9 - 2^8 = 512 + 512 - 256 = 768$$

$\therefore 768$ bit-strings Ans

Ques-4 The
are to be
uppercase
by a po
what is
that can



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auditorium
with an
letter followed
not exceeding 100.
number of chairs
differently?

20

Solve :- There are
and integers 1 through 100 (100 choices)
Every Distinct (letter, number) pair
is a Distinct label.

$$\begin{aligned}\text{Total} &= 26 \times 100 \\ &= 2600\end{aligned}$$

$\therefore$ 2600 Different labels $\underline{\underline{\text{Ans}}}$

Ques-6 If 200 faculty members are
speak french and 50 speak
Russian and 20 can speak both
french and Russian. find how
many faculty members can speak
either french or Russian?

Solve:- Union = 200
Union = 240

formula $\Rightarrow$ french

$\therefore$ 240 faculty



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(21)

french or Russian

Que-7 How many permutations can be made out of the letters of the word 'BASIC'? How many of these

- (i) begin with B?
- (ii) end with C?
- (iii) B and C occupy the end places?

Solve:- (i) Total permutation = $5! = 120$

Begin with B: fix first $\rightarrow 4! = 24$

(ii) End with C: fix last $\rightarrow 4! = 24$

(iii) B and C occupy the ends: two choices which gets B (vs) C, and permute the remaining 3 letters in the middle
 $2 \cdot 3! = 2 \cdot 6 = 12$

$\Rightarrow$ (i) 24 (ii) 24 (iii) 12 Ans

Que-10 How many not divisible by none

Solve :- Count one

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Five integers are divisible and 11? by at least 4 from 1000. (22)

$$① \left[\frac{1000}{3} \right]$$

$$② \left[\frac{1000}{7} \right] = 142$$

$$③ \left[\frac{1000}{11} \right] = 90$$

$$\text{pairs (1cm): } 21 \rightarrow 47, \\ 33 \rightarrow 30, \\ 77 \rightarrow 12$$

$$\text{Sum pairs} = 47 + 30 + 12 \\ = 89$$

$$\text{Triple 1cm } 3 \times 7 \times 11 = 231 \\ 231 \rightarrow \left[\frac{1000}{231} \right] = 4$$

$$\text{So Divisible by } \geq 1 \Rightarrow 333 + 142 + 90 - 89 + 4 \\ \Rightarrow 480$$

$$\text{Thus the divisible by none } \Rightarrow 1000 - 480 \\ \Rightarrow 520$$

$\therefore$ 520 positive integers Ans

Ques-11) Use general
the follow

(i) $a_n - 2a_{n-1} =$

Solve :- Characteristic

$\Rightarrow r^2 - 2r =$

$\Rightarrow (r-3)(r+1) = 0$

$\Rightarrow r = 3, -1$

General $A_n = A \cdot 3^n + B(-1)^n$. Using initial conditions

$n=0 : A+B=1$

$n=1 : 3A-B=3$

Solving $\Rightarrow A=1, B=0$. Hence

$\{A_n = 3^n\} \Rightarrow A_n$

(ii) $y_{n+2} - 4y_{n+1} + 3y_n = 0, y_0=2, y_1=4$

Solve :- Characteristic Equation :-

$\Rightarrow r^2 - 4r + 3 = (r-1)(r-3)$

General $A_n = A + B \cdot 3^n$: from initials,



to solve (23)
a relation :-

$n \geq 2$ with
 $a_0=3, a_1=1$

relation :-

$$2) A + B = 2$$

$$A + 3B = 4$$

Now After solving

$$A = 1, B = 1$$

$$3) \{ A_n = 1 + 3^n \} \text{ Ans } \underline{2}$$

$$(Pv) A_n = B_{n-1} - 16 A_{n-1} \quad \left. \begin{array}{l} = 16 \\ \text{or} = 80 \end{array} \right\}$$

Solve - Characteristic Equation :-

$$2) r^2 - 8r + 16$$

$$2) (r-4)^2 \rightarrow \text{Double root } 4$$

General :- $A_n = (A + Bn)4^n$

2) Use :-

$$n=0 : A = 16$$

$$n=1 : (A+B)4 = 80$$

$$\therefore A+B = 20$$

$$(A=16) \text{ and } (B=4)$$

$$\{ A_n = (16 + 4n)4^n \} \text{ Ans } \underline{2}$$



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(vii) $a_n = 3a_{n-1} + 1, n \geq 1 (a_0 = 1)$

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Solve :- $A_n = 3a_{n-1} + 1, (a_0 = 1)$

homogeneous $(C \cdot 3^n)$

constant particular (C) satisfies

$$C = 3C + 1$$

$$C = -\frac{1}{2}$$

$$\{ A_n = A \cdot 3^n - \frac{1}{2} \}$$

→ Using Initial Conditions :-

$$\text{Use } A_0 = A - \frac{1}{2} = 1 \Rightarrow A = \frac{3}{2}$$

$$\{ A = \frac{3}{2} \}$$

$$\{ A_n = \frac{3}{2} \cdot 3^n - \frac{1}{2} = \frac{3^{n+1} - 1}{2} \} \text{ Ans}$$



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(8x) $a_{n+2} - 2a_{n+1} + a_n = 2^n, (n \geq 0)$

$$\{ a_0 = 2, a_1 = 1 \}$$

Solve - homogeneous part has double root
 $\therefore \rightarrow$ general homogeneous
 $\{ A + Bn \}$

$$\{A_n = A + Bn + 2^n\}$$

(26)

• Use $(n=0)$

$$A+1=2$$

$$\{A=1\}$$

$$\text{Use } n=1: 1+B+2=1$$

$$B=-2$$

$$\{A_n = 1 - 2n + 2^n\} \text{ Ans}$$



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